Entregável 3.3.B: Report for Creating Change Clusters and Conducting Accuracy Assessment for Tile T29TNE

This report reviews the processes used to create visualizations of the breaks detected by the CCDC model and assessing the accuracy of those breaks.

Parameters

Tile Used - T29TNE_0999
Start Date - 2018-09-01
End Date - 2021-10-31
CRS - EPSG: 32629
Date range for grouping pixels together - 10
Minimum polygon area - 0.5 ha
reference data for validation - BDR_DGT_300
Break day tolerance for accuracy assessment- 60

Creating the clusters

Grouping the pixels that experienced a similar break together was done through the script raster_polygon_processing.py. This is a graph based algorithm where the input variables are the locations and chage detection dates for the pixels, the date range tolerance for connecting neighbor pixels (10 days), and the minimum area (0.5 ha) clusters need to cover to be saved to the final output. For this process, the date range was 2018-09-01 to 2021-10-31. This range was selected because it is the most recent dates that overlap with the reference data set that will be used in the accuracy assessment. The selected date range tolerance for validation was 30 days.

Starting with the T29TNE CCDC results, the results were broken into groups of every two months (set-oct 2018, nov-dec 2018, jan-fev 2019, ..., set-out 2021). In each group, the most recent break date was saved for each pixel, and then a raster file where the value of each pixel was the recent break date in the format YYYYMMDD. Next step was to create a graph where nodes are pixels and edges connect pixels that are neighbors (we used 8-connectivity) and have dates that differ at most 10 days. Then, we apply the connected components algorithm fo the graph to get clusters of pixels. Using the pixel values, the mean date for each cluster is computed and assigned to that cluster. Then rasterio's shapes module was used to create polygons of connected pixels with the same assigned date. Finally, all polygons with areas under 0.5 ha were deleted.

The end result of this process were 2 directories, one containing raster files and one containing polygon files for every 2 months between the date ranges.

Accuracy Assessment

The accuracy assessment was done through the script raster_avaliacao_exatidao.py. The input variable theta is the tolerance margin between the predicted break date and the reference break date, and for our assessment it was set to 60 days. It also uses the 2 directories of raster and polygon files of the clusters of areas where breaks occurred.

For every 2 month period, the script first uses the polygon file to create a mask of the raster file, which creates a raster file only containing the pixels that were grouped together with areas largers than 0.5 ha. This new masked raster is then spatially joined with the BDR reference data. The difference between the two dates from the predicted and the reference data is calculated, and the results are classified in the accuracy assessment based on whether or not the difference is within the tolerance margin.

The results table is below, with an average F1 Score of 74.03. Because the assessment was only ran on pixels where a change was detected, there are no True Negative results, which is to be expected.

Results Table

filename	f1_score	omission_error	commision_error	total_VP	total_FP	total_FN	total_VN	had_polygon_mask
output_raster_ccd_20180901_to_20181031.tif	29.92	52.89	78.08	212	755	238	0	True
output_raster_ccd_20181101_to_20181231.tif	42.33	35.54	68.48	156	339	86	0	True
output_raster_ccd_20190101_to_20190228.tif	55.86	23.46	56.02	336	428	103	0	True
output_raster_ccd_20190301_to_20190430.tif	67.53	17.52	42.83	1144	857	243	0	True
output_raster_ccd_20190501_to_20190630.tif	58.29	20.82	53.88	327	382	86	0	True
output_raster_ccd_20190701_to_20190831.tif	95.92	1.49	6.55	5824	408	88	0	True
output_raster_ccd_20190901_to_20191031.tif	85.56	9.51	18.86	2065	480	217	0	True
output_raster_ccd_20191101_to_20191231.tif	62.53	26.2	45.75	338	285	120	0	True
output_raster_ccd_20200101_to_20200229.tif	70.58	17.37	38.4	1075	670	226	0	True
output_raster_ccd_20200301_to_20200430.tif	65.36	20.58	44.47	683	547	177	0	True

filename	f1_score	omission_error	commission_error	total_VP	total_FP	total_FN	total_VN	had_polygon_mask
output_raster_ccd_20200501_to_20200630.tif	76.5	16.34	29.53	1802	755	352	0	True
output_raster_ccd_20200701_to_20200831.tif	97.11	2	3.77	4954	194	101	0	True
output_raster_ccd_20200901_to_20201031.tif	98.16	1.15	2.52	12812	331	149	0	True
output_raster_ccd_20201101_to_20201231.tif	52.01	44.73	50.89	388	402	314	0	True
output_raster_ccd_20210101_to_20210228.tif	54.59	41.83	48.57	972	918	699	0	True
output_raster_ccd_20210301_to_20210430.tif	84.65	8.73	21.07	502	134	48	0	True
output_raster_ccd_20210501_to_20210630.tif	83.83	8.28	22.81	1262	373	114	0	True
output_raster_ccd_20210701_to_20210831.tif	83.53	7.05	24.15	936	298	71	0	True
output_raster_ccd_20210901_to_20211031.tif	59.77	18.76	52.73	849	947	196	0	True
GRAND_TOTAL	**74.03**	**14.64**	**34.65**	**36637**	**19426**	**6284**	**0**	**N/A**